THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2$ equals: 1

(a) 109 **(b)** 81 **(c)** 53 **(d)** 67

2. If x : y = 2 : 3 and y : z = 4 : 7, then x : y : z is: 1

(a) 2 : 4 : 7 **(b)** 8 : 12 : 21 **(c)** 2 : 7 : 7 **(d)** 2 : 3 : 7

3. Two articles are sold at ₹ 1800 each, one at a gain of 10% and the other at a loss of 10%. The overall result is a loss of (in %): 1

(a) 10% **(b)** 5% **(c)** 1% **(d)** 0%

4. The legs of a right triangle are in the ratio $5 : 12$ and its hypotenuse is 52 cm. Its perimeter (in cm) is: 1

(a) 104 **(b)** 68 **(c)** 124 **(d)** 120

5. The present ages of Ria and Sia are in the ratio 3:5. After 8 years the ratio will be 2:3. Ria's present age (years) is: 1

(a) 22 **(b)** 24 **(c)** 40 **(d)** 32

6. If $x - y = 3$ and $xy = 20$, then $x^2 + y^2$ equals: 1

(a) 29 **(b)** 9 **(c)** 49 **(d)** -31

7. The present ages of two brothers are in the ratio 3 : 2. After 4 years the ratio becomes 7 : 5. The present age of the elder brother is: 1

(a) 40 years **(b)** 24 years **(c)** 28 years **(d)** 16 years

8. The population of a town is 50000 and it grows at 10% per annum. After 2 years it will be: 1

(a) 60500 **(b)** 54450 **(c)** 66550 **(d)** 60000

9. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 45 cm. The area of the rectangle (in cm²) is: 1

(a) 405 **(b)** 972 **(c)** 1053 **(d)** 126

10. ₹ 4500 is divided among three friends in the ratio 2:3:5. The difference between the largest and the smallest share (in ₹) is: 1

(a) 1351 **(b)** 900 **(c)** 2250 **(d)** 1350

11. The value of $512^2 - 488^2$ is: 1

(a) 576 **(b)** 500288 **(c)** 24100 **(d)** 24000

12. ₹ 1300 is divided in parts proportional to $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. The largest part is: 1

(a) ₹ 400 **(b)** ₹ 600 **(c)** ₹ 260 **(d)** ₹ 300

13. **Assertion (A):** $(x+5)^2 = x^2 + 10x + 25$. 1

Reason (R): $(a+b)^2 = a^2 + 2ab + b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** The ratios $3 : 4$ and $9 : 12$ are in proportion. 1

Reason (R): Two ratios $a : b$ and $c : d$ are in proportion if they are equal, that is, if $ad = bc$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** 25% of 360 is 90. 1

Reason (R): $x\%$ of a quantity N equals $\frac{x}{100} \times N$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If $x + y = 12$ and $xy = 35$, find $x^2 + y^2$. 2

17. The sides of two squares are in the ratio 3 : 7. Find the ratio of their perimeters and of their areas. 2

18. A number is increased by 25% and the result is decreased by 10%. Is the final number more or less than the original? By what percent? 2

19. The legs of a right triangle are in the ratio $3 : 4$ and the hypotenuse is 35 cm. Find the legs. 2

20. If $x : y = 3 : 4$, find the ratio $(5x + 2y) : (4x - 1y)$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. If $x - y = 4$ and $xy = 21$, find $x^2 + y^2$ and $(x + y)^2$. 3

22. The ratio of the present ages of a father and son is 3 : 1. After 4 years the ratio will be 7 : 3. Find their present ages. 3

23. A merchant sells two items at ₹ 1980 each, one at a gain of 10% and the other at a loss of 10%. Find the overall gain or loss in rupees. 3

24. A ladder 37 m long, with its foot in a street, reaches a window 35 m high on one side. Without moving its foot it is turned to reach a window 12 m high on the opposite side. Find the width of the street. 3

OR

A garrison of 240 men has provisions for 45 days. After 15 days 60 men leave. For how many more days will the remaining provisions last for the men who stay? 3

25. Simplify $(3x + 2)^2 - (3x - 2)^2$. 3

OR

15 men can finish a job in 24 days. After 8 days, 5 more men join them. In how many more days is the job finished? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Case study. A rectangular vegetable plot has length $(x + 5)$ m and breadth $(x + 2)$ m. A square flower bed beside it has side $(x + 3)$ m. 5

(i) Write the area of the vegetable plot as an expression.

(ii) Find its area when $x = 10$.

(iii) Write the perimeter of the plot.

(iv) By how much does the area of the plot exceed that of the flower bed? Verify for $x = 10$.

27. A sum of ₹ 10000 is invested for 3 years at 20% p.a. compounded annually. Find the interest for each of the three years separately and verify that their total is the compound interest. 5

OR

In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 9 cm, BC = 12 cm, CD = 20 cm and DA = 25 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

END OF QUESTION PAPER — Total: 50 Marks

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